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Column Assigned: V82

Time Series Analysis

DATA.STAT.780

Optional Bonus: TimesFM Forecast (Compare with ARIMA)

I used TimesFM 2.5 version in a zero-shot setting to generate forecasts for $h = 10, 25$, and 100 steps ahead using the same train/test split as ARIMA(3,1,1) trained on observations 1–100 and evaluated against observations 101–200.

MSE Comparison Table

Model	h=10	h=25	h=100
ARIMA(3,1,1)	1.97	3.84	10.64
TimesFM 2.5 (zero-shot)	2.30	5.52	17.31
Winner	ARIMA	ARIMA	ARIMA

ARIMA(3,1,1) outperforms TimesFM 2.5 across all three forecast horizons. The gap widens significantly at longer horizons — at $h=100$, TimesFM's MSE of 17.31 is nearly 63% higher than ARIMA's 10.64, indicating that TimesFM struggles substantially in the long run.

From the plot, both models fail to capture the oscillatory behaviour of the test series after observation 100. However ARIMA(3,1,1) performs better numerically. Neither model captures the key turning points in the test data.

Conclusion: ARIMA(3,1,1) is the superior model for this particular series across all horizons. This result is not entirely surprising, TimesFM is a general-purpose foundation model trained on diverse datasets

